MBG 1201 Introduction to Modern Biology

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Text book

Campbell Biology, 12th Edition.
Urry, Cain, Wasserman, Minorsky, Reece

Outline

What is biology?
What is evolution?
The properties of life
5 Unifying themes of biology
Classifying the diversity of life
The 3 domains of life
Scientific method

What is biology?

- **Biology is the scientific study of life.**
- Broad field due to the wide variety of life.
- Its scope includes Molecular biology, biochemistry, genetics, cell biology, microbiology, immunology, neurobiology, ecology, systems biology etc.

- **Evolution** : a process of biological change in which species accumulate differences from their ancestors as they adapt to different environments over time.



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ALX1 (P/P)



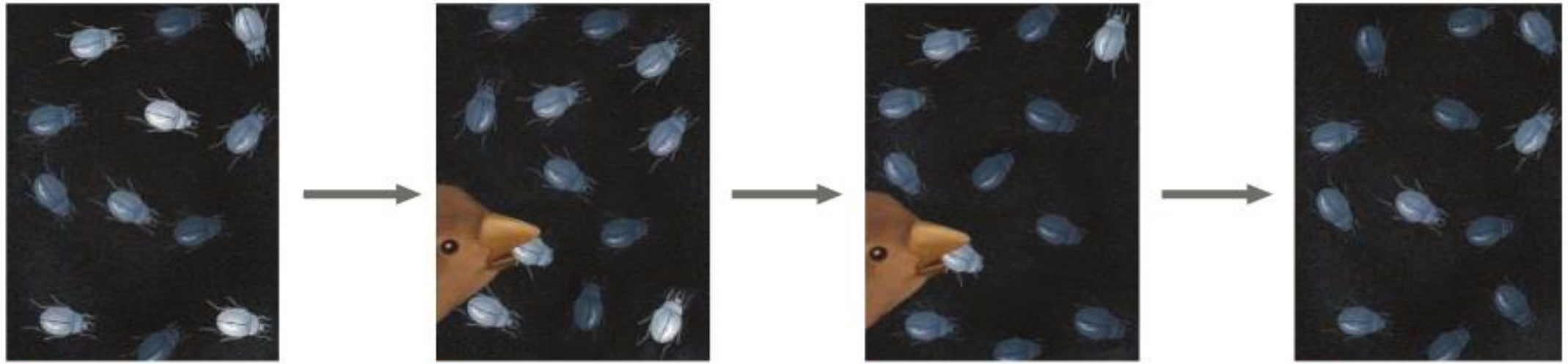
HMGA2 (L/L)
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HMGA2 (L/L)
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Natural Selection

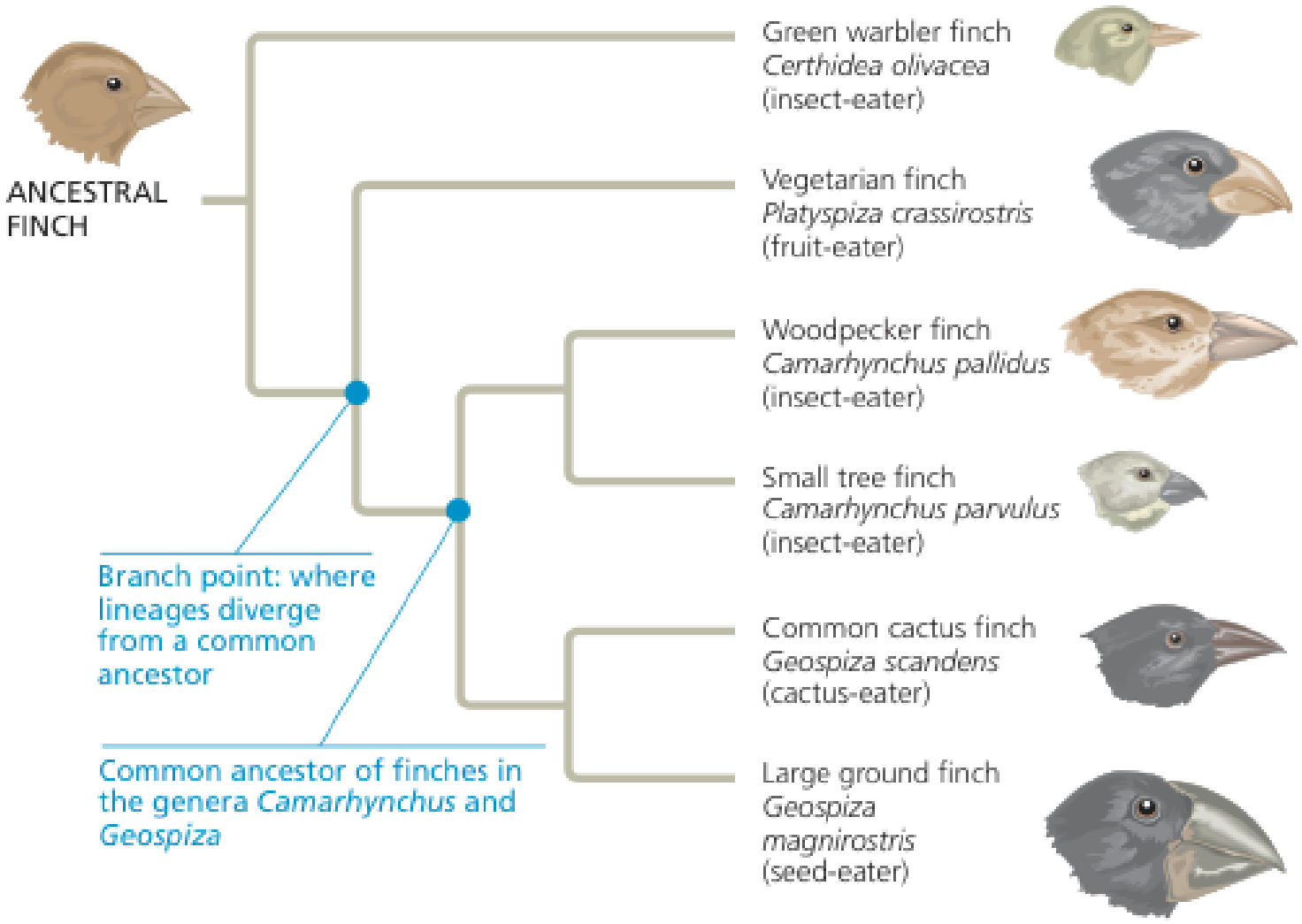
▼ **Figure 1.18 Natural selection.** This imaginary beetle population has colonized a locale where the soil has been blackened by a recent brush fire. Initially, the population varies extensively in the inherited coloration of the individuals, from very light gray to charcoal. For hungry birds that prey on the beetles, it is easiest to spot the beetles that are lightest in color.



1. Population with varied inherited traits
2. Elimination of individuals with certain traits that make them more visible
3. Reproduction of survivors
4. Increased frequency of traits that enhance survival and reproductive success

▶ Figure 1.20 Descent with modification: adaptive radiation of finches on the Galápagos Islands. This "tree" illustrates a current hypothesis for the evolutionary relationships of finches on the Galápagos. Note the various beaks, which are adapted to particular food sources. For example, heavier, thicker beaks are better at cracking seeds, while the more slender beaks are better at grasping insects.

Mastering Biology HHMI Video: The Origin of Species: The Beak of the Finch



Properties of life

All living things exhibit
Growth and development: Get bigger and more complex. Cells enlarge and divide; differentiate.

Energy Metabolism: The chemical reactions of living things; nutrients are taken and utilized as sources of energy (catabolism) and as sources to build up cell components (anabolism).

Homeostasis: maintain a relatively controlled internal environment. In achieving homeostasis, organisms maintain a “steady state”—a relatively constant internal environment—even when the external environment changes significantly.
pH, temperature, chemical concentrations should be maintained in a narrow range.

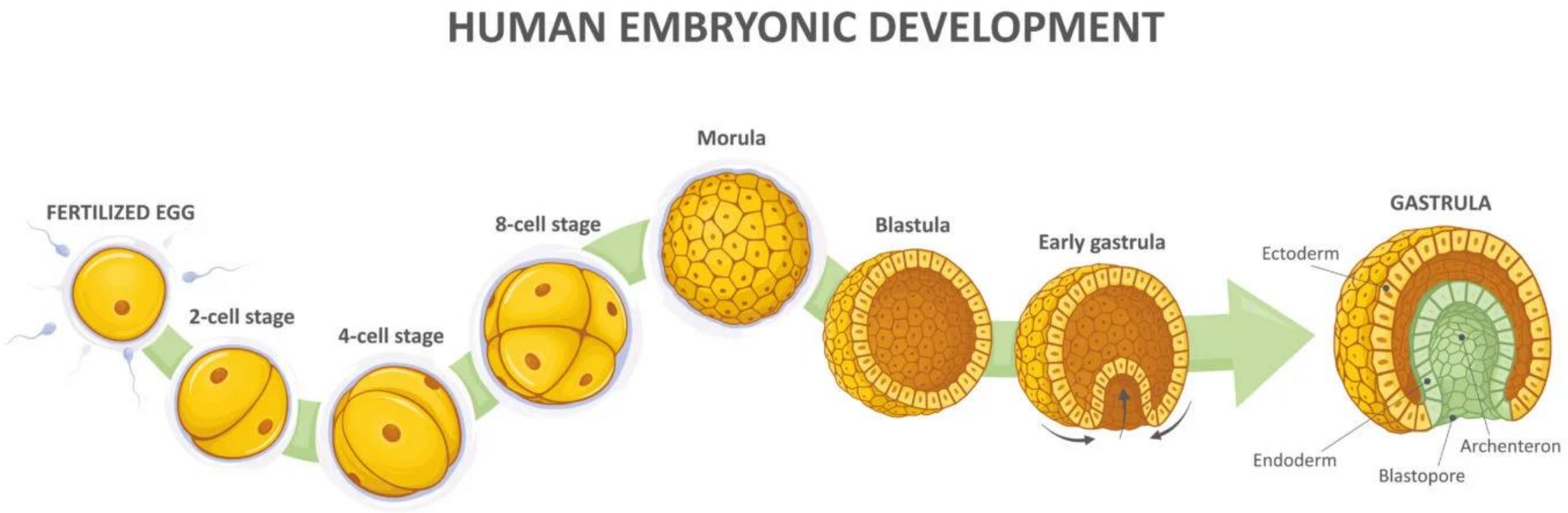
Adaptation: living things adjust over time due to mutation and natural selection which improves survival.

Response to stimuli: make changes in response to external stimuli. To maintain homeostasis and to survive.
e.g. Chemotaxis, phototaxis / can be positive or negative

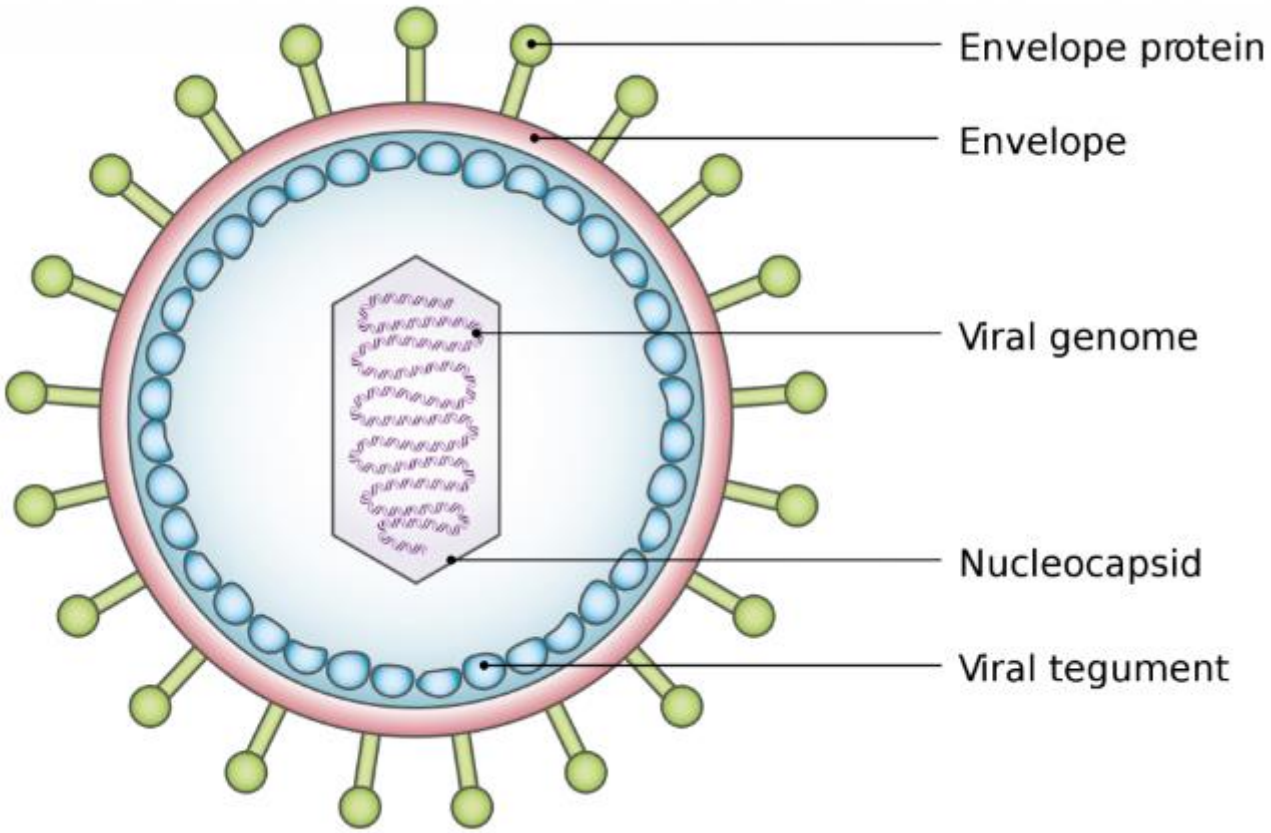
Organization: Living things are made from organic compounds (carbon-based); they are composed of cells and are either unicellular or multi-cellular. Proteins with well defined shapes, Organelles, cellular compartments, tissues, organs, organ systems.

Reproduction: Produce offspring which carry hereditary traits of the parent organism.

Properties of life

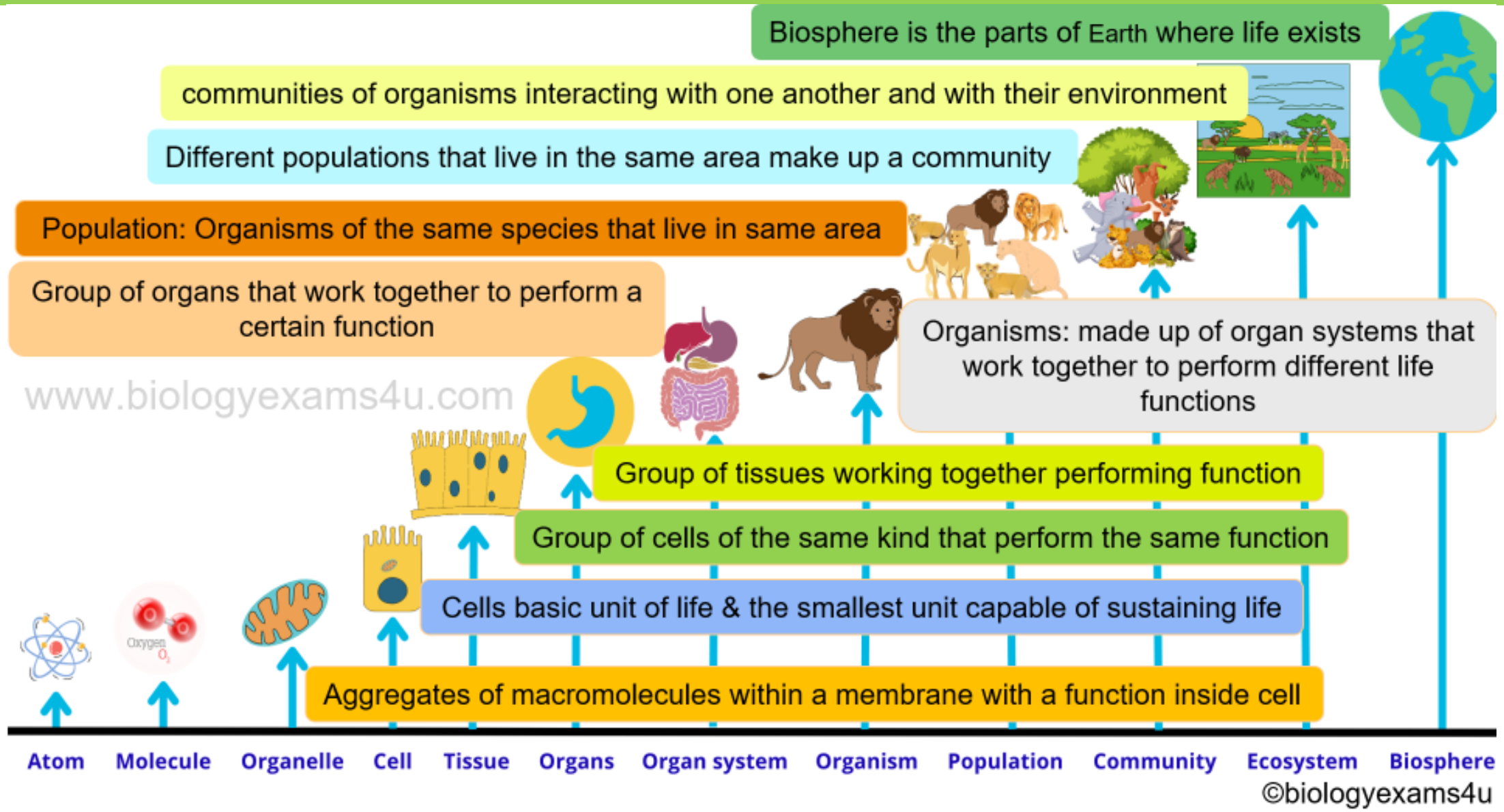


Properties of life

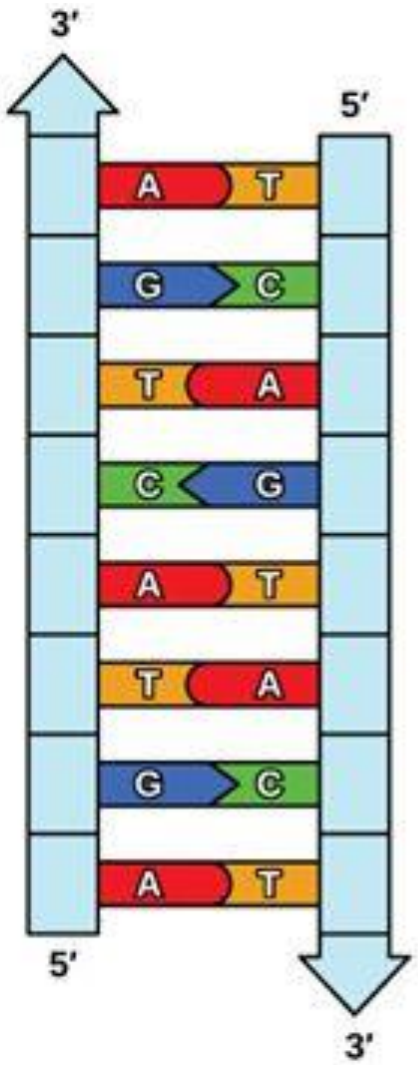


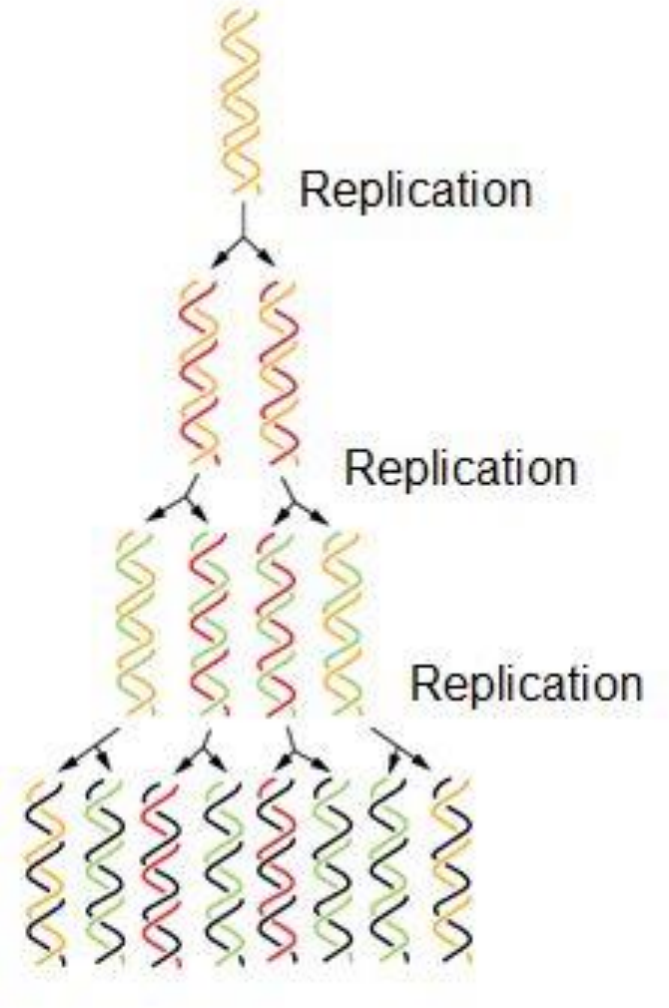
Are viruses alive?
Growth and development
Energy metabolism
Homeostasis
Adaptation
Response to stimuli
Organization
Reproduction

5 Unifying Themes of Biology: Organization



5 Unifying Themes of Biology: Information





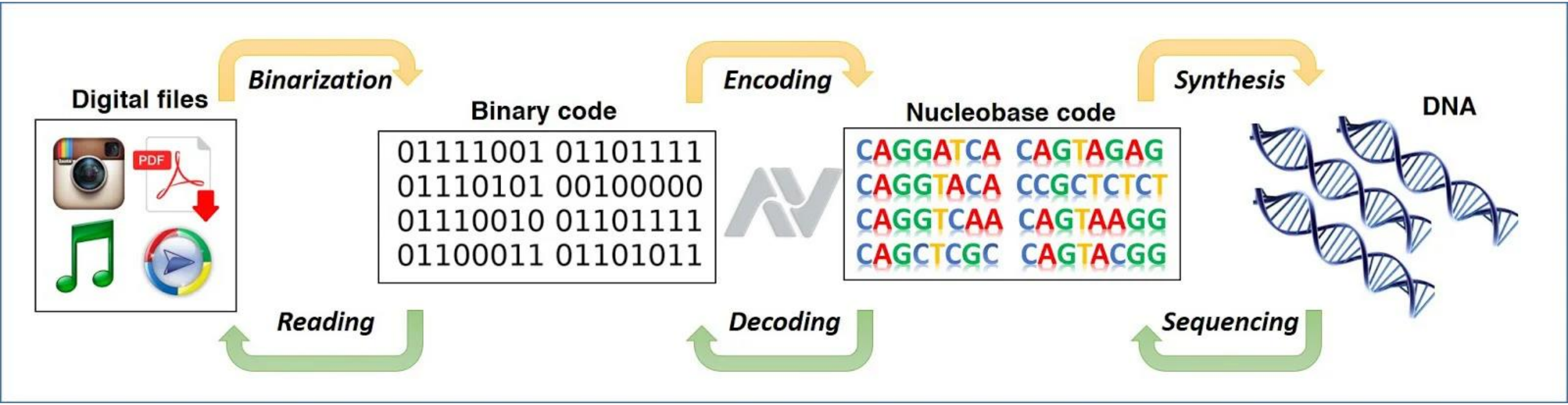
Deoxyribonucleic acid, or DNA, is the special molecule that sets the living world apart from the nonliving by carrying the hereditary instructions that allow organisms to grow and reproduce.

DNA contains information for building proteins that are vital for cell growth and reproduction.

Each DNA molecule is made up of two long chains arranged in a double helix.

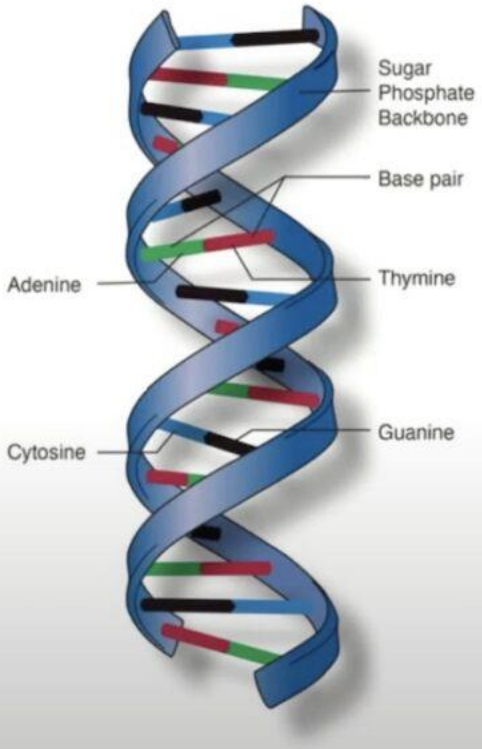
Each link of a chain is one of four kinds of chemical building blocks called nucleotides.

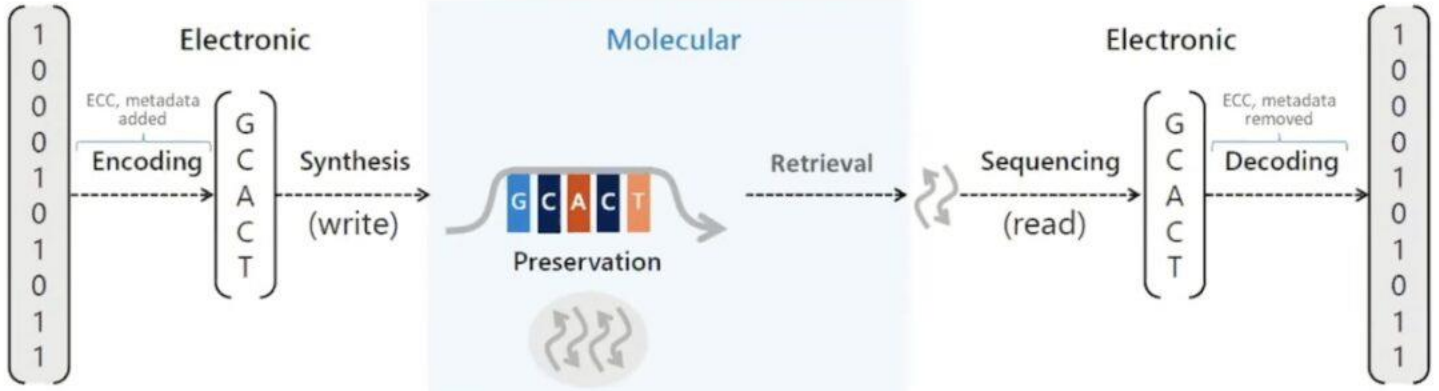
5 Unifying Themes of Biology: Information



5 Unifying Themes of Biology: Information

DNA and the DNA Data Storage Pipeline





Basis of what makes DNA data storage possible

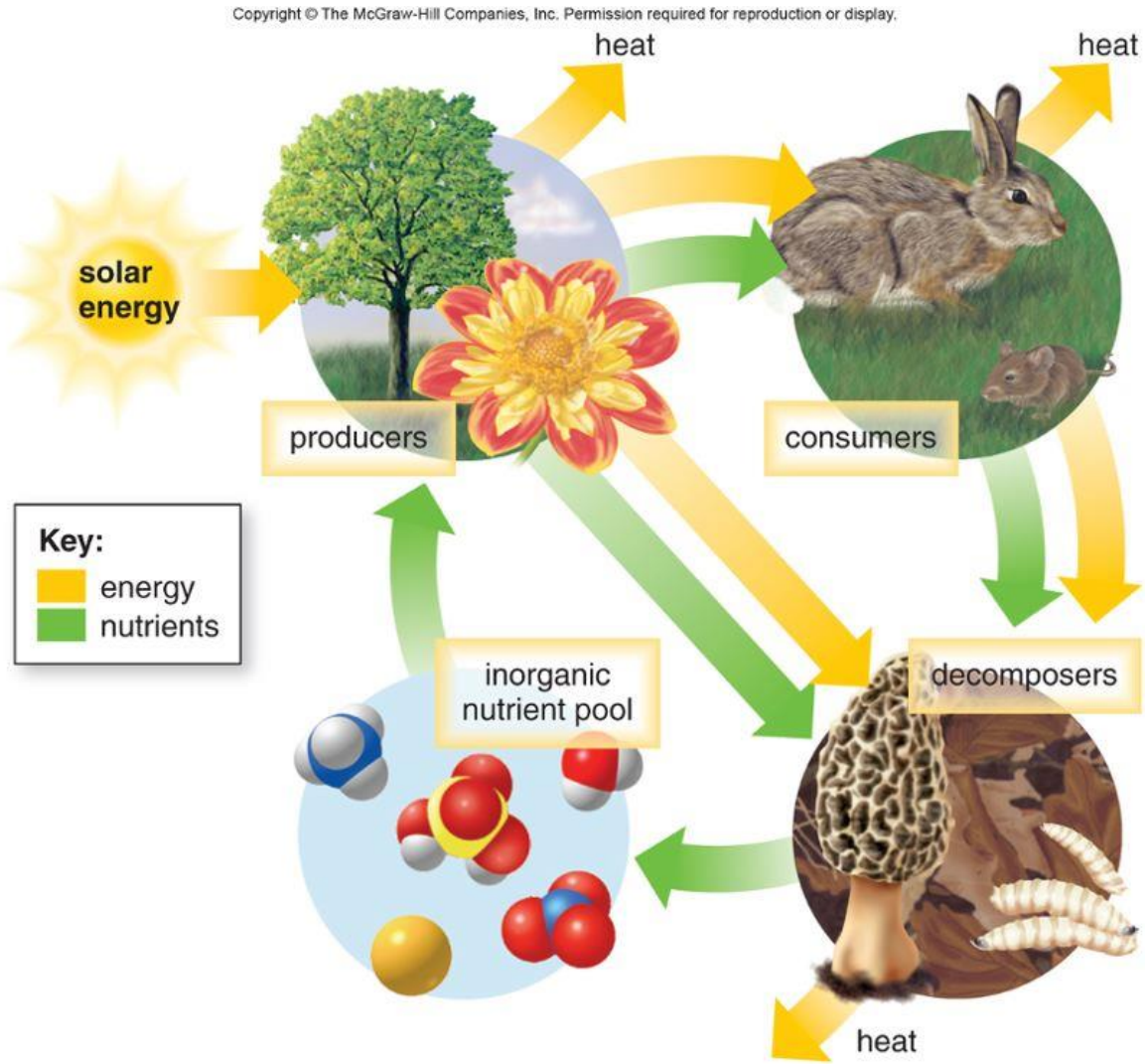
- Information in DNA is encoded in four bases (AGCT)
- We can today construct synthetic DNA molecules, base-by-base
- Hence, we can now encode digital bits in DNA molecules

DNA DATA STORAGE ALLIANCE

5 Unifying Themes of Biology: Energy and matter

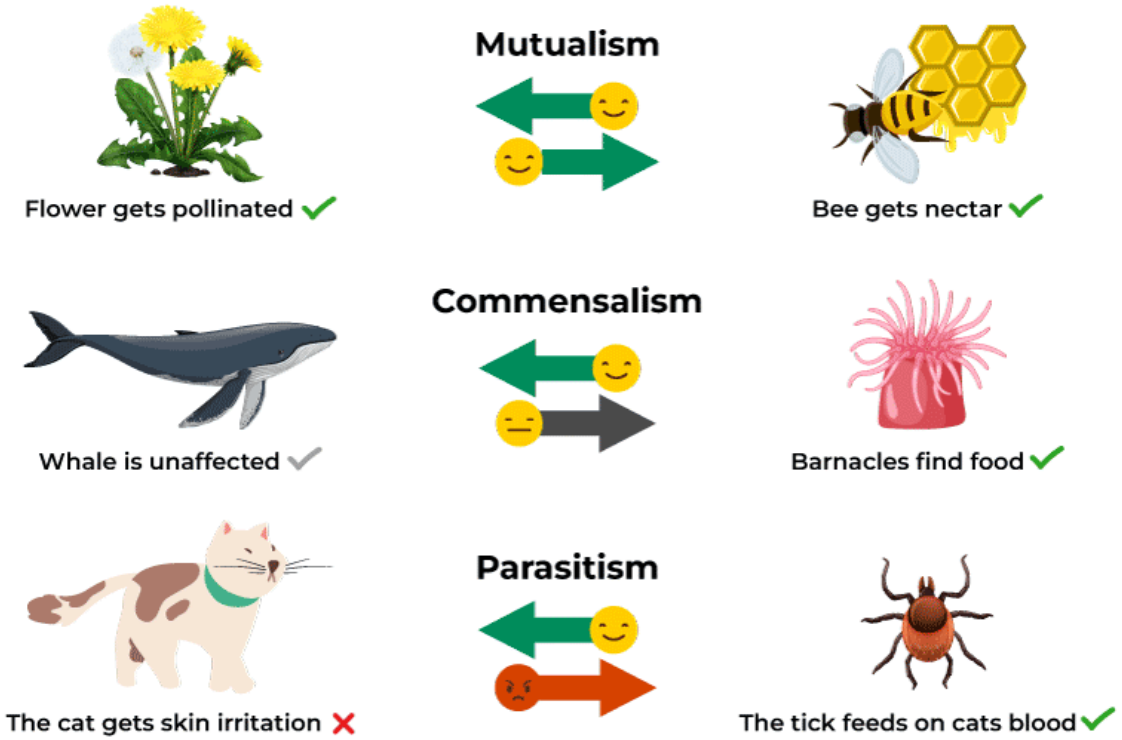
All living things use energy to carry out life's activities.

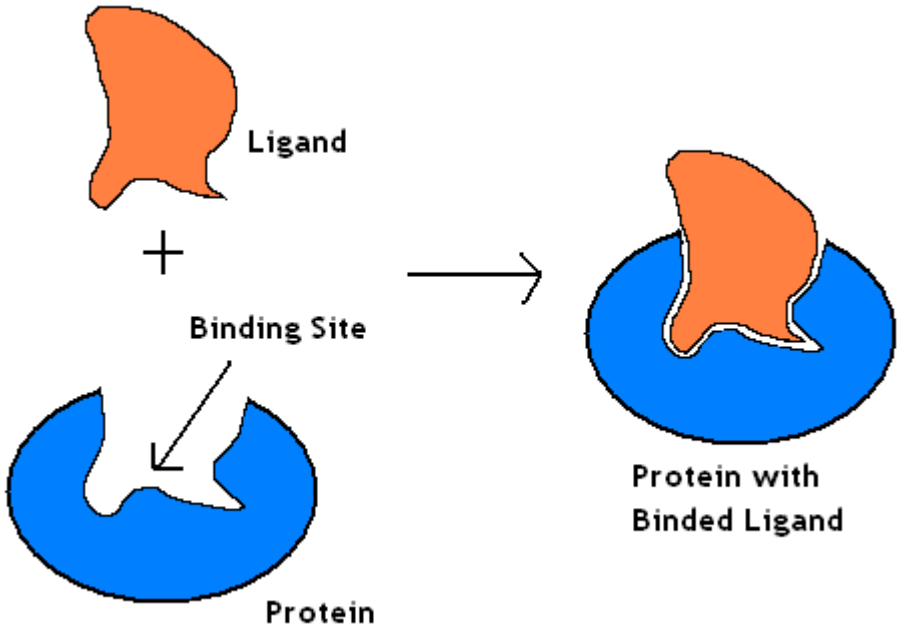
Moving, growing, reproducing, and the various cellular activities of life are work.
Work requires energy.
Ongoing inputs of energy and nutrients sustain all organisms, as well as nature's overall organization.



5 Unifying Themes of Biology: Interaction

Interaction occurs in all levels.

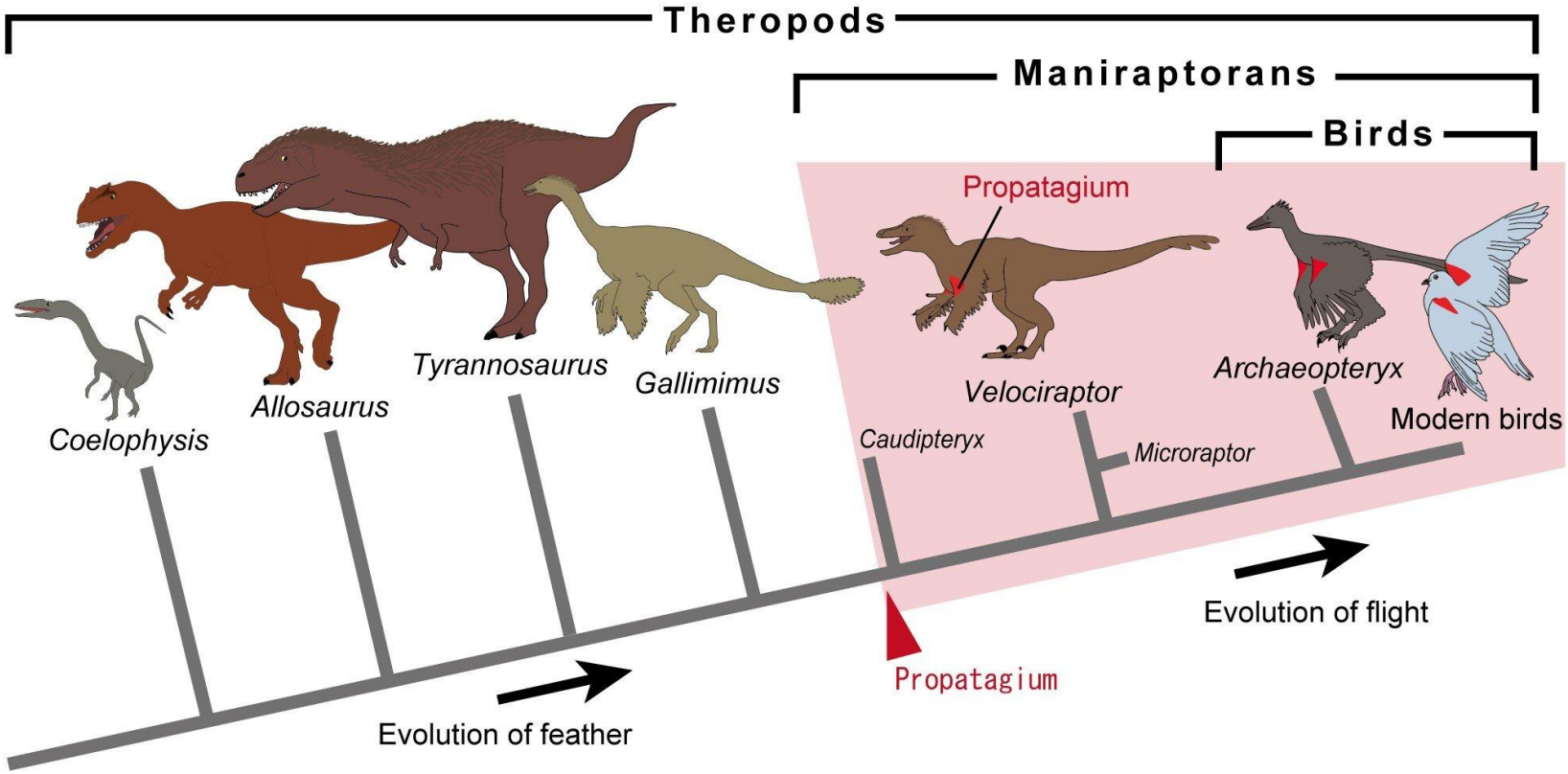




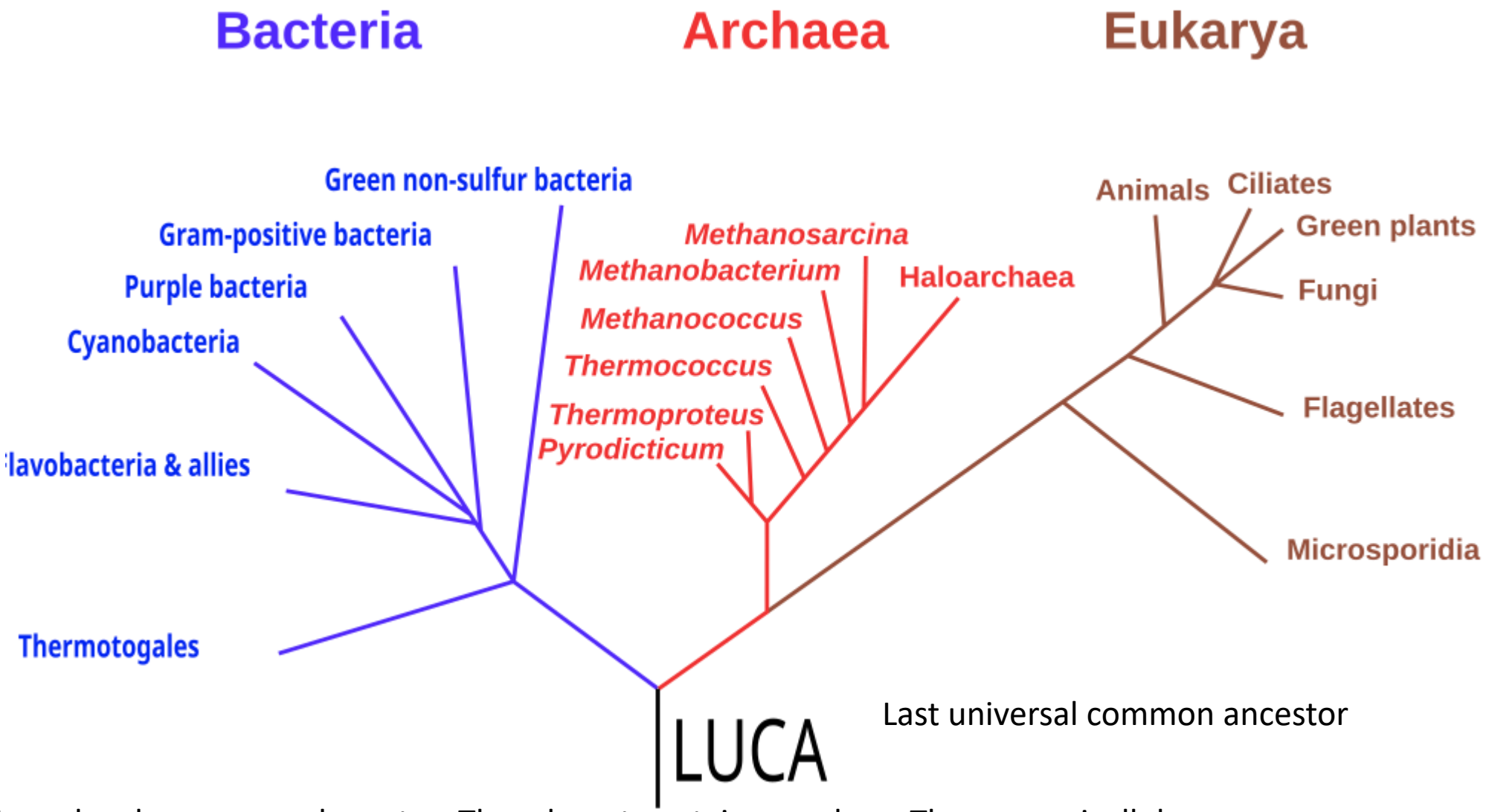
5 Unifying Themes of Biology: Evolution

The organisms living on earth today are the modified descendents of common ancestors.

evolution: a process of biological change in which species accumulate differences from their ancestors as they adapt to different environments over time.



Biologists arrange the diversity of life into three domains



Bacteria and archaea are prokaryotes. They do not contain a nucleus. They are unicellular.

Biologists arrange the diversity of life into three domains



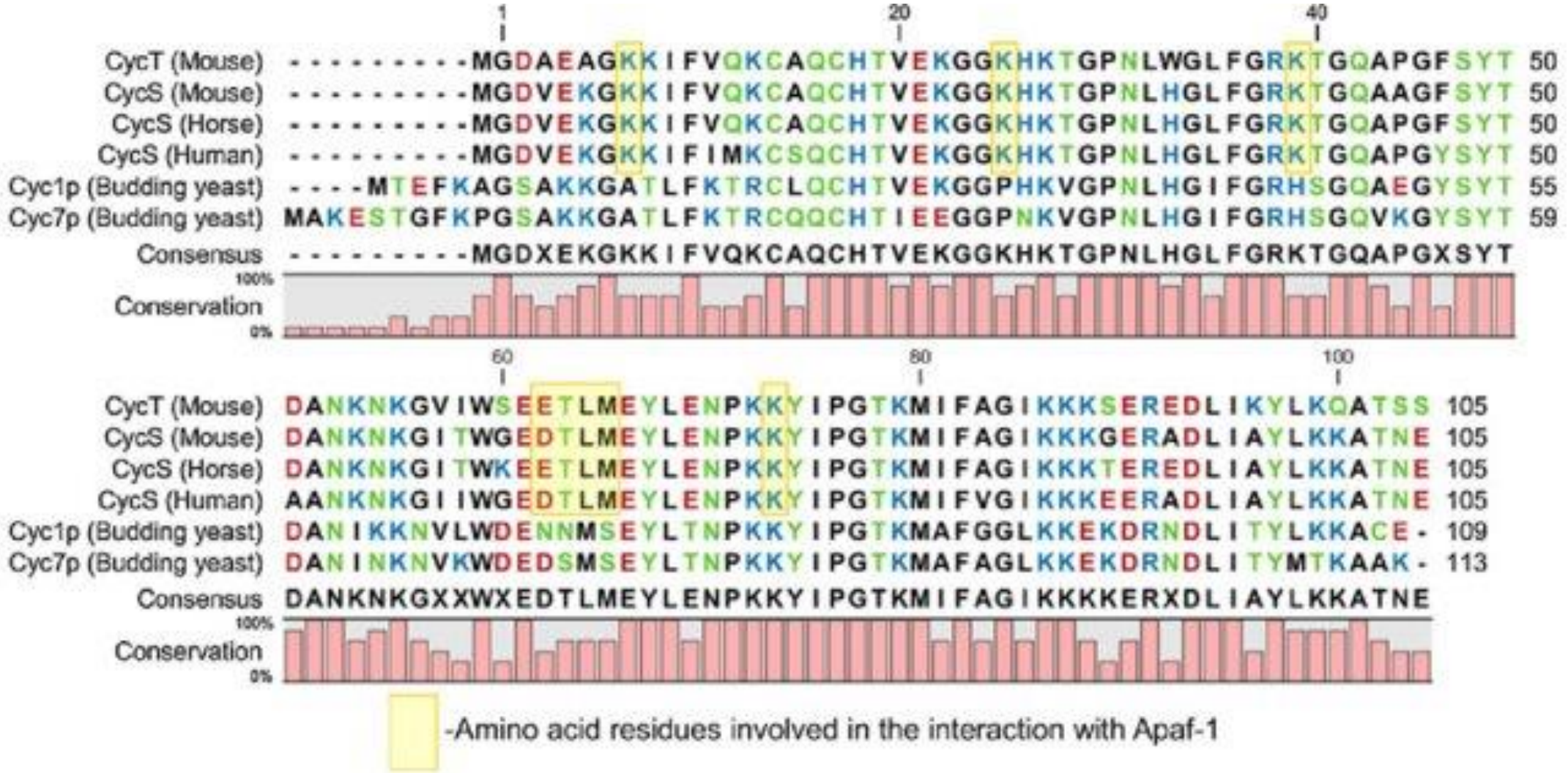
The three major divisions (domains) of the living world.

TABLE 1–2 Some Model Organisms and Their Genomes

Organism	Genome size* (nucleotide pairs)	Approximate number of genes
Escherichia coli (bacterium)	4.6×10^6	4300
Saccharomyces cerevisiae (yeast)	13×10^6	6600
Caenorhabditis elegans (roundworm)	130×10^6	21,000
Arabidopsis thaliana (plant)	220×10^6	29,000
Drosophila melanogaster (fruit fly)	200×10^6	15,000
Danio rerio (zebrafish)	1400×10^6	32,000
Mus musculus (mouse)	2800×10^6	30,000
Homo sapiens (human)	3200×10^6	30,000

*Genome size includes an estimate for the amount of highly repeated DNA sequence not in genome databases.

Unity in the Diversity of Life



IF SO MUCH UNITY, WHY SO MANY SPECIES?

Charles Darwin published *On the Origin of Species by Means of Natural Selection* in 1859

Darwin made two main points:

- Species showed evidence of “descent with modification” from common ancestors (as species adapt to different environments over time, they accumulate differences from their ancestors.)
- Natural selection is the mechanism behind “descent with modification”

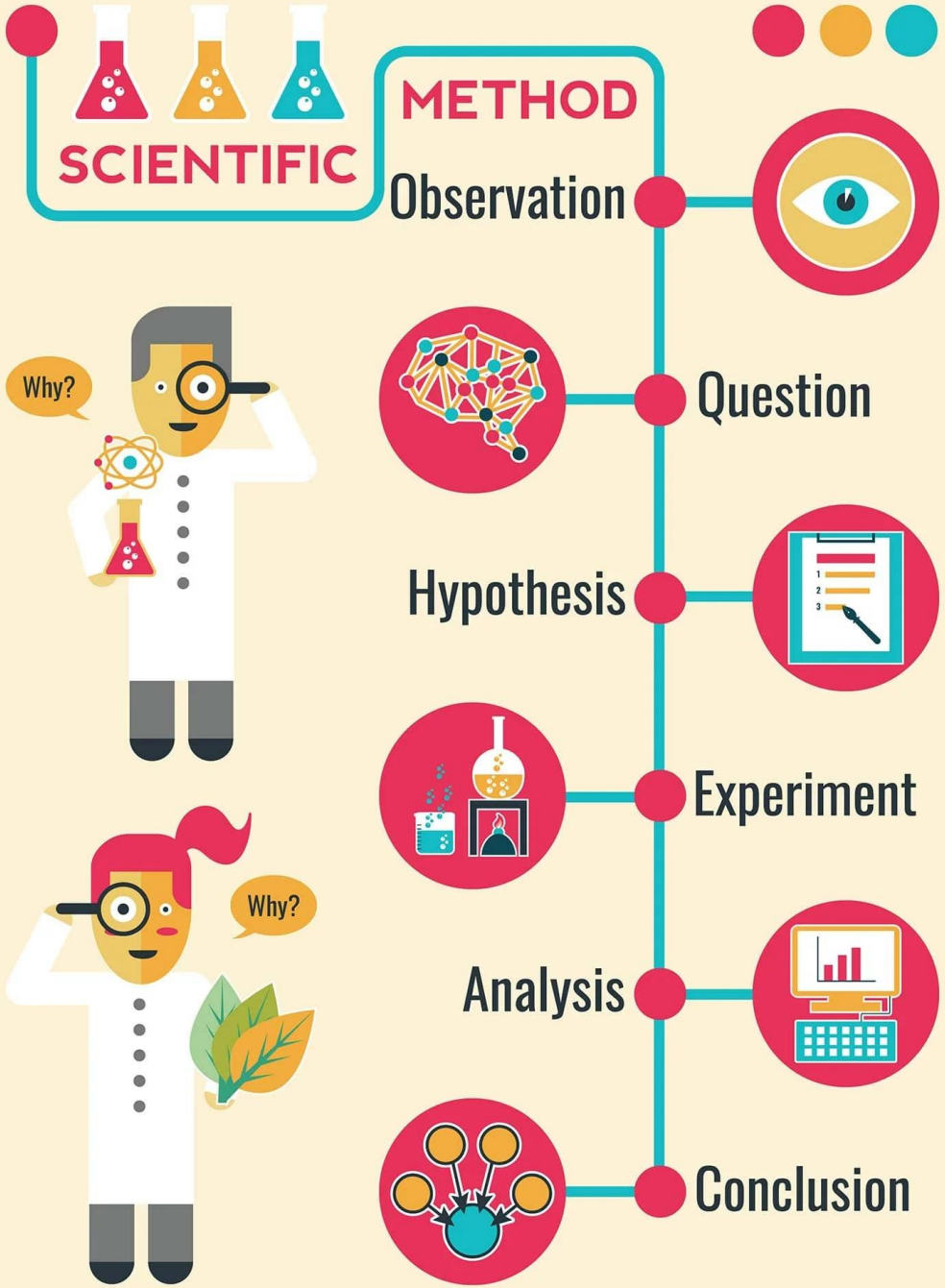
Darwin’s theory explained the duality of unity and diversity

Summary of Darwin's Theory

Darwin's theory of evolution can be summarized as follows:

- Individual organisms differ, and some of this **variation can be inherited**.
- Organisms produce **more offspring than can survive**, so there is **competition for limited resources**.
- Individuals **best suited to their environment survive and reproduce** most successfully.
- The most "fit" organisms **pass traits to their offspring**. This process of **natural selection** causes species to **change over time.**

On The Origin Of Species By Means Of Natural Selection (1859)



İnquiry: the search for information and explanations of natural phenomena.

Scientists use a process of inquiry that includes making observations, forming logical, testable explanations (hypotheses), and testing them.

The process is necessarily repetitive: In testing a hypothesis, more observations may inspire revision of the original hypothesis or formation of a new one, thus leading to further testing.

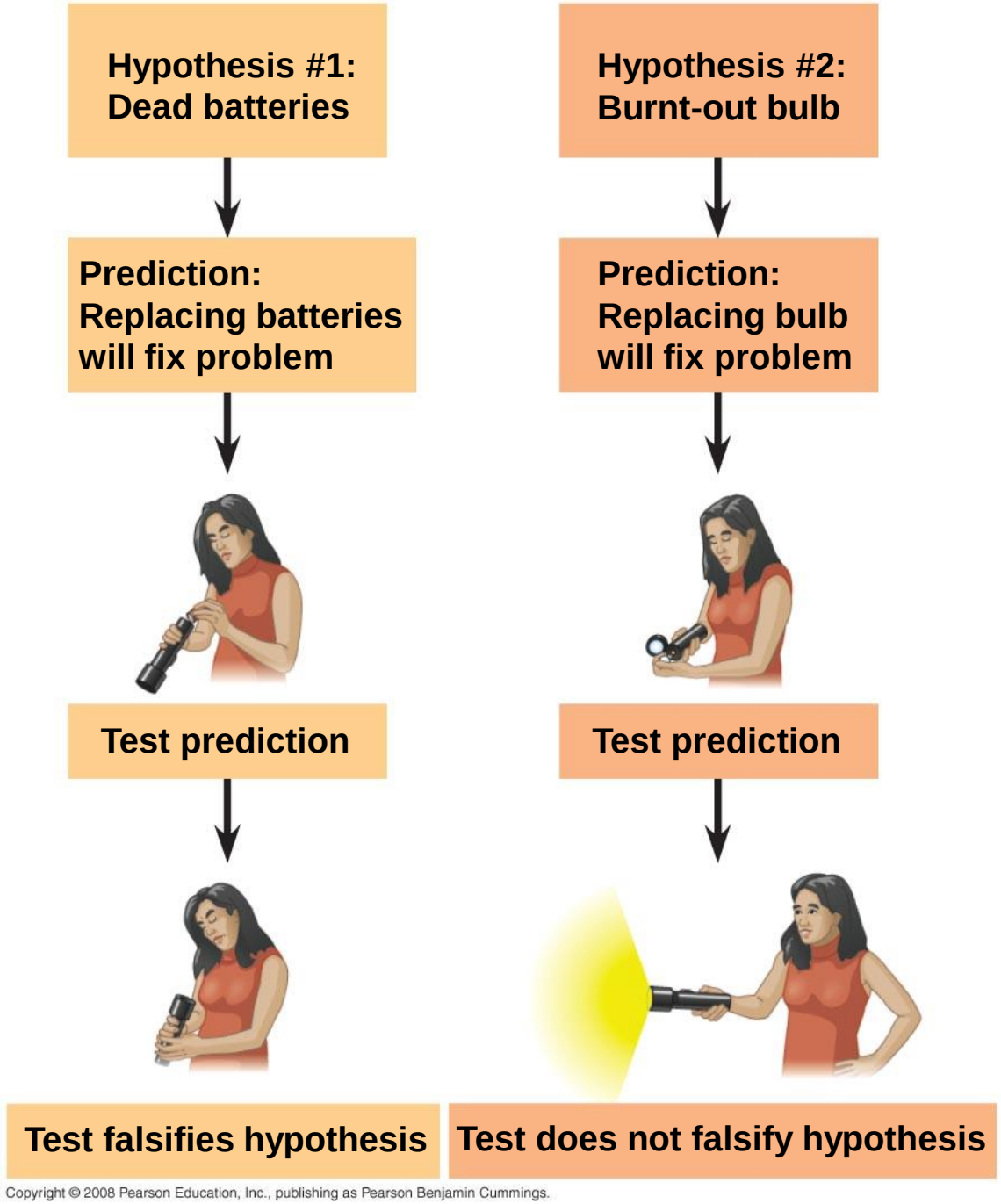
A hypothesis is a tentative answer to a well formed question.

THE NATURE OF BIOLOGICAL INQUIRY

Table 1.3 Example of a Scientific Approach

1.	Observation	People get cancer.
2.	Question	Why do people get cancer?
3.	Hypothesis	Smoking cigarettes may cause cancer.
4.	Prediction	If smoking causes cancer, then individuals who smoke will get cancer more often than those who do not.
5.	Gather information	Conduct a survey of individuals who smoke and individuals who do not smoke. Determine which group has the highest incidence of cancers.
	Laboratory experiment	Establish identical groups of laboratory rats (the model system). Expose one group to cigarette smoke. Compare the incidence of new cancers in each of the two groups.
6.	Assess results	Compile test results and draw conclusions from them.
7.	Report	Submit the results and the conclusions to the scientific community for review and publication.

Fig. 1-24b



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THE NATURE OF BIOLOGICAL INQUIRY

- A hypothesis must be *testable* and *falsifiable*
- Hypothesis-based science often makes use of two or more alternative hypotheses
- Failure to falsify a hypothesis does not *prove* that hypothesis For example, you replace your flashlight bulb, and it now works; this supports the hypothesis that your bulb was burnt out, but does not prove it (perhaps the first bulb was inserted incorrectly)

THE NATURE OF BIOLOGICAL INQUIRY

About the Word "Theory"

1. A scientific theory has been tested many different times and ways in the natural world and has not yet been disproved.

a. Theories are accepted or rejected on the basis of tests, and are subject to revision.

b. For example: Darwin's theory of evolution by natural selection answers many diverse questions about the natural world successfully and yet is still being tested.

2. The fact that an idea, or even a theory, might be subject to change indicates the strength of science, not weakness.